

Harshkumar Rajubhai Panchal

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EDUCATION

B.Tech in AI & Data Science – SCET, Surat

2022–2026

HSC – V.S. Patel School (60%)

SSC – V.S. Patel School (74.17%)

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience in Machine Learning, Deep Learning, and Computer Vision. Proven ability to build intelligent, scalable applications from medical imaging to real-time sign language translation with a strong foundation in Python, TensorFlow, and full-stack development.

EXPERIENCE

Research Intern – SVNIT Surat

Jan 2026 – June 2026 (6 months)

Deep Learning, Computer Vision, NLP, TensorFlow/PyTorch

- Completed a 4-month research internship under the guidance of Dr. Praveen Chandaliya and Dr. Pruthwik Mishra at SVNIT Surat, focused on real-time Indian Sign Language (ISL) Video-to-Text Translation.
- Utilized MediaPipe Holistic for landmark extraction and implemented continuous sign language recognition techniques to handle sentence-level datasets.
- Designed and trained deep learning architectures (including a custom Multi-Stream Spatial-Temporal Transformer) on GPU infrastructure for gesture sequence modeling.

PROJECTS

Forest Fire Detection using Satellite Imagery

2025

Python, TensorFlow/Keras, CNN | GitHub Link

- Designed and trained a CNN model for forest fire detection from satellite imagery.
- Improved early detection capability by automating analysis of large-scale image datasets.

Diamond Price Predictor

2025

Flask, XGBoost, JavaScript | GitHub Link

- Implemented an XGBoost-based prediction system for diamond pricing.
- Designed responsive UI with real-time input validation and API-based predictions.

Milky Music Bot Website

2024

React, Vite, JavaScript | GitHub Link

- Engineered a responsive frontend using React and modular components.
- Optimized performance using Vite for faster build and rendering.

Diabetic Retinopathy Detection

2025

CNN, Deep Learning | GitHub Link

- Trained a CNN for diabetic retinopathy detection from medical images.
- Applied data preprocessing and augmentation techniques to improve model accuracy.

TECHNICAL SKILLS

- Programming: Python, C, SQL
- AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision
- Libraries: TensorFlow, PyTorch, Scikit-learn, OpenCV, NumPy, Pandas
- Visualization: Matplotlib, Seaborn
- Web Development: HTML, CSS, JavaScript, React, Flask

CERTIFICATIONS

- Tata Data Analytics Job Simulation – Forage (June 2025)
- AWS APAC Solutions Architecture Virtual Experience – Forage (July 2025)
- Frontend & Backend using JS and React Workshop – GDSC, MLSA, Devtown
- Python Workshop – MLSA, Devtown

ACTIVITIES

- Volunteer – NSS, SCET
- Volunteer – Converse Tech Fest (2023, 2024)
- Participated in theatre workshops and performed in group drama events

LANGUAGES

English, Hindi, Gujarati

INTERESTS

Sports, Acting, Gaming, Leadership, Learning Emerging Technologies